PPGBG Dataset Documentation

Dataset name: PPGBG
Version: v1.0
Release folder: `PPGBG_v1.0_release/`
Application contact: PPGBGa@163.com
Citation requirement: Works using PPGBG must cite the associated publication. PPGBG does not have a separate dataset DOI in this release.

This document defines the contents, structure, variables, protocol summary, permitted research context, and known limitations of PPGBG. Access to and use of PPGBG are governed by the PPGBG Data Use Agreement.

1. Overview

PPGBG is a longitudinal, subject-independent benchmark dataset for non-invasive blood glucose estimation from fingertip photoplethysmography (PPG). It includes synchronized cuff blood pressure measurements and clinical metadata. The dataset supports repeated-measurement encounter-level evaluation, distribution-drift analysis, incremental or continual learning research, and secondary cuffless blood pressure analysis.

Item	Value
Subjects	183 (112 male, 71 female)
Clinical encounters	528
Valid 40 s PPG acquisitions	523
Validated 4 s PPG windows	5,224
Signal	Fingertip PPG, 1000 Hz, 4 s windows (4000 samples)
Reference BG	Capillary blood glucose (mmol/L)
Blood pressure	Cuff SBP/DBP (mmHg)
Collection span	285 days
Label modality	Per-encounter clinical measurements and per-subject demographics/ comorbidities

2. Files in this Archive

```
PPGBG_v1.0_release/
|-- data/
|   |-- signals.npy                # float32 array, shape (5224, 4000); PPG windows
```

```
|   |-- windows.csv                # 5224 rows; one row per PPG window
|   |-- encounters.csv             # 528 rows; one row per clinical encounter
|   `-- subjects.csv               # 183 rows; one row per subject
|-- documents/
|   |-- en/                        # English PDF documents
|   `-- zh/                        # Chinese PDF documents
`-- SHA256SUMS.txt                 # checksums for release files
```

Signal-table correspondence: row `i` of `data/signals.npy` corresponds to row `i` of `data/windows.csv`. The `window_id`, `subject_id`, and `encounter_id` fields provide the indexing relationship between windows, clinical encounters, and subjects.

3. Signal Format

- **Shape:** `(5224, 4000)`.
- **Data type:** `float32`.
- **Sampling rate:** 1000 Hz.
- **Window duration:** 4 s per row, corresponding to 4000 samples per window.
- **Segmentation:** non-overlapping 4 s windows obtained from valid 40 s PPG acquisitions.
- **Pre-processing:** each 4 s window was band-pass filtered with a zero-phase, forward-backward, fourth-order Butterworth filter with a 0.5-8 Hz passband. Each window was then z-score normalized using its own mean and standard deviation. The released array is already filtered and standardized at the window level.
- **Additional processing:** no additional baseline correction, motion-artifact suppression, wavelet transform, or detrending is included in the released array.
- **Acquisition-to-window relation:** a valid 40 s acquisition normally produces ten 4 s windows. Six valid acquisitions contain nine windows because one short-duration flat-line window was removed during curation.

```
import numpy as np
import pandas as pd

X = np.load("data/signals.npy")
W = pd.read_csv("data/windows.csv")
y = W["bg"].to_numpy()
```

4. Label Dictionary

4.1 windows.csv

Column	Type	Unit/ coding	Description
`window_id`	string	-	Unique window identifier.
`subject_id`	string	-	Subject identifier.
`encounter_id`	string	-	Clinical encounter identifier.
`window_index`	integer	-	Window position within the corresponding encounter.

valid_40s	integer	0/1	Hardware-conformity indicator for the source 40 s acquisition; all rows in windows.csv have value 1.
sex	string	M/F	Sex.
age	integer	years	Age at encounter.
bmi	float	kg/m^2	Body mass index.
weight	float	kg	Weight.
height	float	m	Height.
hr	integer	bpm	Heart rate measured by the reference monitor.
sbp, dbp	integer	mmHg	Systolic and diastolic blood pressure.
bg	float	mmol/L	Reference capillary blood glucose target.
hypertension, diabetes, cvd, ckd	integer	0/1	Comorbidity indicators.
active_infection, active_fever, active_postoperative, active_other, active_none	integer	0/1	Acute condition indicators at encounter level.
smoking_never, smoking_former, smoking_current	integer	0/1	Smoking status indicators.

4.2 encounters.csv

encounter_id, subject_id, encounter_index, clinical_encounter_count, effective_window_count, valid_40s, hr, sbp, dbp, bg.

Each row represents one clinical encounter. effective_window_count reports the number of retained 4 s windows for that encounter. In this release, 517 encounters have 10 windows, 6 encounters have 9 windows, and 5 encounters have valid_40s = 0 with 0 retained windows. Rows with 0 retained windows are retained for longitudinal encounter accounting but do not correspond to rows in signals.npy or windows.csv.

4.3 subjects.csv

subject_id, model_group, model_diabetic_group, clinical_encounters, valid_40s_acquisitions, final_valid_4s_windows, sex, weight_kg, height_m, age_years, bmi_kg_m2, hypertension, diabetes, cvd, ckd, active_*, smoking_*.

model_group and model_diabetic_group define the analysis stratification used in the associated publication: 39 subjects are assigned to the Diabetic analysis group and 144 subjects to the Non-diabetic analysis group. This modeling split is not identical to the clinical diabetes comorbidity flag, which is positive for 57 subjects.

5. Cohort Statistics

Subject-level demographic variables are calculated from `subjects.csv`. HR, SBP, DBP, and BG are calculated at encounter level from `encounters.csv`.

Metric	Mean	Min	Max	SD
Weight (kg)	69.76	46.30	98.00	10.93
Height (m)	1.69	1.55	1.87	0.07
Age (years)	57.02	21	96	21.20
BMI (kg/m^2)	24.35	16.80	33.51	3.20
HR (bpm)	72.98	55	102	7.67
SBP (mmHg)	130.02	85	164	18.57
DBP (mmHg)	71.74	44	94	9.85
BG (mmol/L)	6.56	4.2	16.3	2.12

Baseline comorbidities: hypertension 55%, diabetes 31%, cardiovascular disease 43%, chronic kidney disease 8%. Smoking status: never 43%, former 26%, current 31%.

6. Acquisition Protocol Summary

Data were collected under an open-environment protocol. Participants were seated with the arm resting. No fasting or medication constraints were imposed, so that intra-individual variation and practical measurement conditions were retained. Each session lasted approximately 10 min and included a 5 min acclimatization period. Reference capillary BG and PPG were measured from the left index fingertip. Blood pressure was measured on the right upper arm. Measurements within an encounter were completed within an approximately 2 min window.

- Reference BG: Accu-Chek Performa (Roche), measurable range 0.6-33.3 mmol/L, daily L1/L2 quality control.
- Blood pressure: Omron J751 upper-arm monitor.
- PPG: EVAL-ADPD4100Z-PPG board and EVAL-ADPDUCZ MCU (Analog Devices), 1 kHz sampling.

7. Curation Pipeline

The candidate cohort contained 195 subjects. The final release contains 183 subjects and 5,224 retained 4 s PPG windows.

1. Integrity screening removed 3 subjects with invalid or out-of-range labels.
2. Availability screening removed 2 subjects with sampling-rate or duration violations.
3. Skewness-based signal-quality screening removed 7 subjects below the cohort-mean signal-quality threshold.
4. The remaining valid 40 s acquisitions were segmented into non-overlapping 4 s windows.
5. Short-duration flat-line screening removed 6 individual windows.

Final release: 183 subjects, 528 clinical encounters, 523 valid 40 s acquisitions, and 5,224 validated 4 s windows.

8. Ethics and Privacy

The study was reviewed and approved by the institutional ethics committee of the participating clinical site. All participants provided informed consent before data collection. The release contains no direct identifiers. Subjects are represented only by pseudonymous identifiers. Re-identification attempts are prohibited under the PPGBG Data Use Agreement.

9. Intended Use and Supported Tasks

- Primary task: non-invasive blood glucose estimation from PPG under subject-independent evaluation.
- Secondary tasks: cuffless blood pressure estimation; incremental or continual learning under longitudinal distribution drift; encounter-level evaluation beyond single-window metrics.
- Appropriate use context: non-commercial academic research.
- Excluded use context: clinical diagnosis, treatment, or real-patient medical decision-making.

10. Known Limitations

- The cohort is single-site and may not generalize to all populations, devices, or clinical environments.
- Reference labels are device measurements rather than laboratory venous blood assays.
- The released PPG signal has already been filtered and window-standardized; raw ADC records are not distributed in v1.0.
- Exact timestamps are not released, so day-level intervals cannot be reconstructed from the release files.

11. Copyright, License, and Citation

Works using PPGBG must cite the associated publication:

> Zhang, Z., Li, J., Wang, Y., Yang, K., and Lei, T. (2026). Dynamic Incremental Learning for Non-invasive Blood Glucose Estimation from Wearable Physiological Signals. Manuscript under review. Preprint DOI: https://doi.org/10.5281/zenodo.21202402

```
@unpublished{zhang2026dil,
  title  = {Dynamic Incremental Learning for Non-invasive Blood Glucose Estimation from Wearable
Physiological Signals},
  author = {Zhang, Zexing and Li, Jichao and Wang, Yilong and Yang, Kewei and Lei, Tianyang},
  year   = {2026},
```

```
  doi    = {10.5281/zenodo.21202402},
  note   = {Manuscript under review; preprint on Zenodo}
}
```

12. Access

To request access, complete and sign `documents/en/DATA_USE_AGREEMENT.pdf` or `documents/zh/DATA_USE_AGREEMENT_zh.pdf`, then send the signed agreement to PPGBGa@163.com.